



5.

As the graph shows, 其殘差呈 Normal distribution
(Which graph 呈 - 直線)

2. 圖形呈喇叭狀, 誤差的 Variance
隨預測值增加而變大

→ 不符合殘差 Variance 的一致性 Assumption

8.

a) Linearity: $E(\varepsilon) = 0$, Independence: 誤差項之間相互獨立

Normality: $\varepsilon \Rightarrow$ Normal, Equal Variance: 對 all 誤差項
的 Variance are Equal

b) σ^2 的 unbiased 估計值 = MSE = 58.05

c) Normal Probability Plot: 所有點連線趨於直線
→ 殘差分配 $\Rightarrow$ Normal

Versus Fits: Points 呈 randomly distributed, 符合 Variance - 一致性

→ Regression model's assumption 有效

9.

a) $r^2 = 0.027 \rightarrow$ Price and Screen size 的 linear relationship 非常小

b) Normal, 符合 Variance-一致性

c) 不符合 Linearity.

12.

$$\bar{X} = \frac{2 \times 5 + 3 \times 5 + 4 \times 5}{15}$$

$$\bar{y} = 8.8$$

$$S_{xx} = 10$$

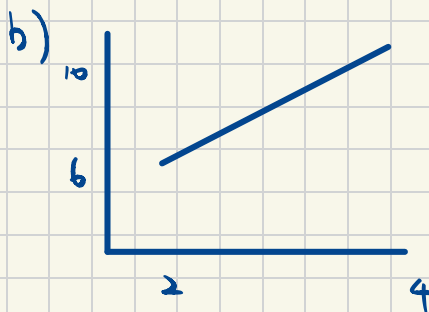
$$S_{yy} = 42.4$$

$$S_{xy} = 15$$

a) $b = 1.5$

$$a = 8.8 - 1.5 \times 3 = 4.3$$

$$\hat{y} = 4.3 + 1.5x$$



c) $SSE = TSS - SSR$

$$= 42.4 - \frac{15^2}{10} = 19.9$$

$$MSE = 19.9 \div 13 \approx 1.5308$$

d) $H_0: \beta = 0$, $H_a: \beta \neq 0$

$$t_{STAT} = \frac{1.5}{\sqrt{\frac{1.5308}{10}}} \approx 3.83$$

$t_{0.05, 13} = 2.16 \rightarrow$ Reject H_0

e) $p\text{-value} = 0.00207 < 0.05 \Rightarrow \text{Reject } H_0$

g) $\hat{y} = 1.5 \times 3 + 4.3 = 8.8$

$$95\% \text{ CI} = b \pm t_{13, 0.025} \times \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{XX}}}$$

$$= 8.8 \pm 2.160 \sqrt{1.53 \left(\frac{1}{15} + \frac{(3-3)^2}{10} \right)} = [8.11, 9.49]$$

15)

a) $\hat{y} = 2 - 0.875x$, $\bar{x} = 0$, $\bar{y} = 2$

$MSE = 0.0833$, $S_{XX} = 16$

$$99\% \text{ CI} = \hat{y} \pm t_{3, 0.005} \times \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{XX}} \right)}$$

$$= 2.875 \pm 5.841 \times \sqrt{0.0833 \left(\frac{1}{5} + \frac{(-1)^2}{16} \right)}$$

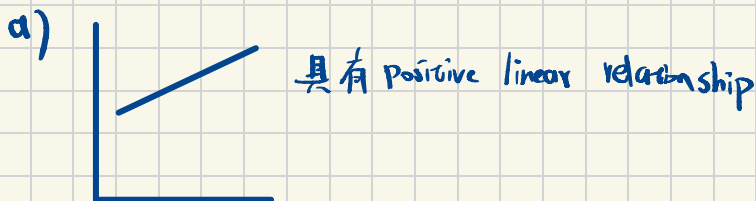
$$= [2.011, 3.739]$$

b) $99\% \text{ PI} = \hat{y} \pm t_{3, 0.005} \sqrt{MSE \left(1 + \frac{(x_0 - \bar{x})^2}{S_{XX}} \right)}$

$$= [-0.769, 3.019]$$

$= \bar{x}$ 時, 預測值 $x_0 = \bar{x} = 0$, PI 最小

18.



b) $\hat{y} = 80.9 + 270.8x$

c) $p\text{-value} = 0.003 < 0.05 \Rightarrow \text{Reject } H_0 \Rightarrow x, y$ 之間存在顯著 linear relationship

d) 90% CI = $\hat{\beta} \pm t_{10, \alpha} \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{XX}} \right)}$
 $= [112.130, 157.9]$

13.

$H_0: \rho = 0, H_a: \rho > 0$

$S_{XX} = 602222.22$

$r = 0.3297$

$S_{YY} = 8200$

$t = \frac{0.3297 \times \sqrt{9-2}}{\sqrt{1-0.3297^2}}$

$\approx 0.924 < t_{7, 0.05} = 1.895$

$\Rightarrow$ Non-reject H_0

$S_{XY} = 13166.67$

14.

a) use correlation analysis.

b) $S_{XX} = 108.857$

$r = 0.919 \Rightarrow$ 幾乎完全正相關

$S_{YY} = 13971.429$

15.

a) $S_{XX} = 10.689$

$S_{YY} = 410.667$

$r = 0.174$

$S_{XY} = 11.533$

b) $H_0: \rho = 0, H_a: \rho \neq 0$

$t = \frac{0.174 \sqrt{12-2}}{\sqrt{1-0.174^2}}$

$\approx 0.559 < t_{10, 0.05} = 2.228$

$\Rightarrow$ Non-reject H_0